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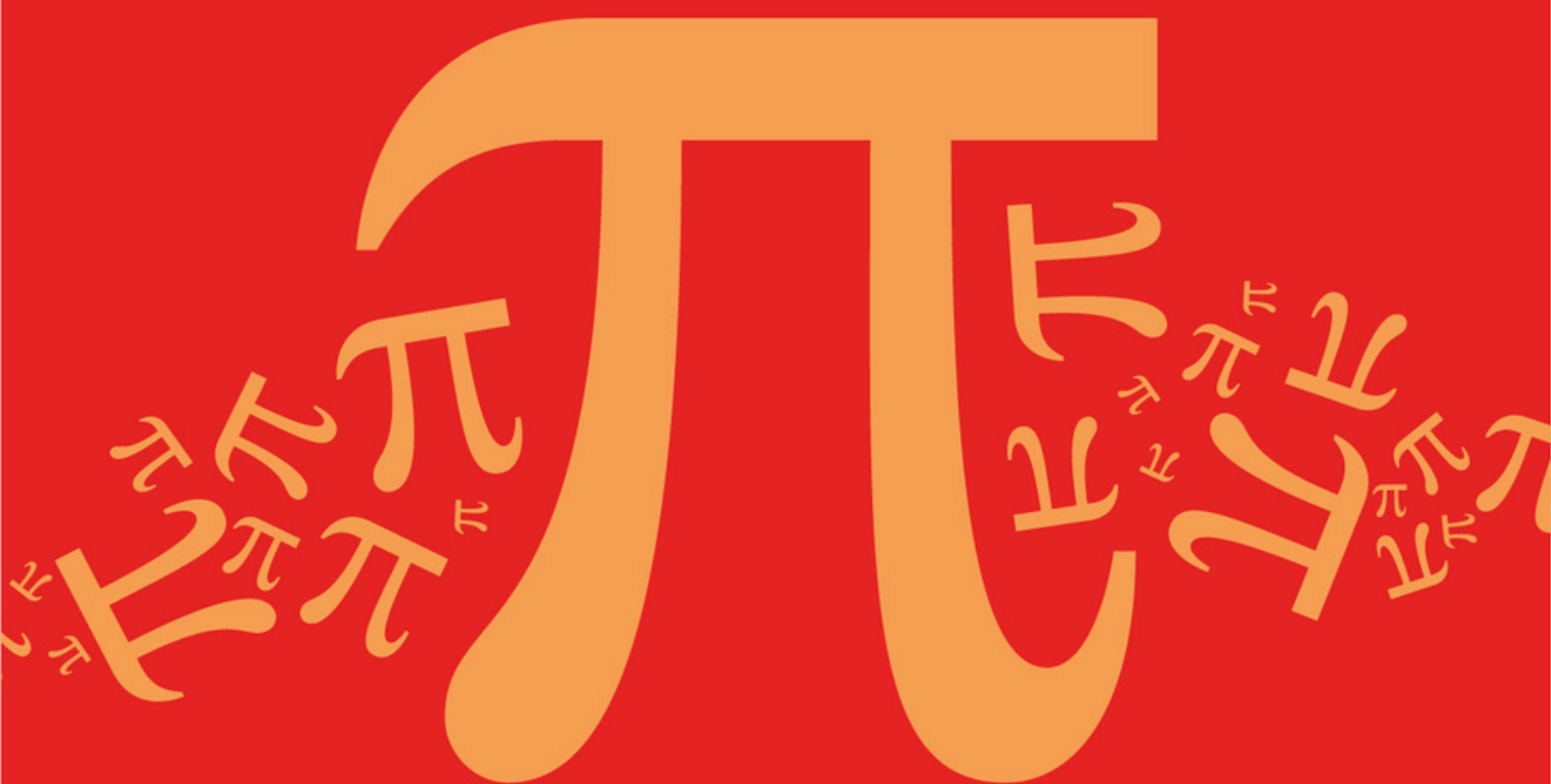
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C6: Singular Matrices and Inverse of a 2x2 Matrix

Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses factorisation or pre-multiplication to isolate A	AO3.1a	M1	$\mathbf{AB} - 2\mathbf{A} = \mathbf{I}$ $\mathbf{A}(\mathbf{B} - 2\mathbf{I}) = \mathbf{I}$
Deduces A in terms of B and I . Could be implied by sight of $\begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}$ with attempt to invert.	AO2.2a	A1	$\mathbf{A} = (\mathbf{B} - 2\mathbf{I})^{-1}$
Obtains correct matrix A .	AO1.1b	A1	$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}^{-1}$ $\mathbf{A} = \frac{1}{-2} \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$

ALT

Marking Instructions	AO	Marks	Typical Solution
Sets up four equations with at least three correct.	AO3.1a	M1	Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2 \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $3a - 4b = 1 + 2a$ and $3c - 4d = 0 + 2c$ and $-2a + 8b = 0 + 2b$ and $-2c + 8d = 1 + 2d$
Deduces at least two correct elements of A . Note: Two correct elements from just two correct equations can score M1A1.	AO2.2a	A1	$a = 4b + 1$ and $c = 4d$ $6b = 2a$ and $6d = 2c + 1$ $\therefore 3b = 4b + 1$ and $6d = 2(4d) + 1$ $-1 = b$ and $-1 = 2d \Rightarrow d = -\frac{1}{2}$ $\therefore a = 4 \times -1 + 1$ and $c = 4 \times -\frac{1}{2}$ $a = -3$ and $c = -2$
Obtains correct matrix A	AO1.1b	A1	$\therefore \mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$
Total 3 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds $k = 2$	AO1.1b	B1	$k - 2 = 0 \Rightarrow k = 2$
(b)	States correct transformation	AO1.2	B1	Reflection in the y-axis
(c) (i)	Finds product BA	AO1.1a	M1	$\mathbf{BA} = \begin{bmatrix} -1 & -2 \\ 1 & k \end{bmatrix}$
	Allow one slip			
	Obtains inverse FT 'their' BA provided M1 awarded	AO1.1b	A1F	$(\mathbf{BA})^{-1} = \frac{1}{-k - (-2)} \begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix}$ $\mathbf{A}^{-1} = \frac{1}{k-2} \begin{bmatrix} k & -2 \\ -1 & 1 \end{bmatrix}$
	Finds $\mathbf{A}^{-1}$ and $\mathbf{B}^{-1}$	AO1.1b	B1	$\mathbf{B}^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{(k-2) \times (-1)}$ $\begin{bmatrix} k+0 & (-2 \times -1)+0 \\ (-1 \times 1)+0 & 0+(-1 \times 1) \end{bmatrix}$
	Obtains correct $\mathbf{A}^{-1}$, $\mathbf{B}^{-1}$, and shows that $(k-2) \times (-1) = 2-k = -k - (-2)$ thus completing verification Must clearly show $\mathbf{A}^{-1} \times \mathbf{B}^{-1}$, method for this mark – disallow if answer simply stated	AO2.1	R1	$\mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{(k-2) \times (-1)}$ $\begin{bmatrix} k+0 & (-2 \times -1)+0 \\ (-1 \times 1)+0 & 0+(-1 \times 1) \end{bmatrix}$ That is the same as $(\mathbf{BA})^{-1}$ $(\mathbf{BA})^{-1} = \frac{1}{2-k} \begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix} = \mathbf{A}^{-1}\mathbf{B}^{-1}$
(ii)	Uses equation for identity from definition	AO3.1a	M1	We require to demonstrate that:
	Commences argument by manipulating the matrix products within the equation with clear pairing	AO2.1	R1	$(\mathbf{NM}) \times \{\mathbf{M}^{-1}\mathbf{N}^{-1}\} = \mathbf{I}$ $(\mathbf{NM}) \times \mathbf{M}^{-1}\mathbf{N}^{-1} = \mathbf{N}(\mathbf{M} \times \mathbf{M}^{-1})\mathbf{N}^{-1}$
	Clearly demonstrates that $\mathbf{M} \times \mathbf{M}^{-1} = \mathbf{I}$ used	AO2.4	B1	$= \mathbf{N} \mathbf{I} \mathbf{N}^{-1}$ $= \mathbf{N} \mathbf{N}^{-1}$
	Completes the argument using rigorous reasoning with definition of matrix inverse and associativity mentioned Must see all working with correct pairing of each matrix with inverse	AO2.1	R1	$= \mathbf{I}$ Using definition of matrix inverse and associativity of matrix multiplication Hence true for all non-singular matrices N and M
				Total 10 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Commences a proof by correctly setting up an equation using the definition of an invariant point	AO2.1	R1	For an invariant point $\mathbf{M} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$
Pre-multiplies by $\mathbf{M}^{-1}$	AO2.1	R1	Pre-multiply both sides by $\mathbf{M}^{-1}$ $\mathbf{M}^{-1} \mathbf{M} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{M}^{-1} \begin{pmatrix} x \\ y \end{pmatrix}$
Uses $\mathbf{M}^{-1}\mathbf{M} = \mathbf{I}$ and concludes their rigorous mathematical argument to deduce that (x, y) is invariant under S AG	AO2.2a	R1	$\mathbf{M}^{-1}\mathbf{M} = \mathbf{I} \text{ hence}$ $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{M}^{-1} \begin{pmatrix} x \\ y \end{pmatrix}$ Therefore $\begin{pmatrix} x \\ y \end{pmatrix}$ is invariant under S.
			Total 3 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains that the claim is incorrect as singular square matrices do not have inverses.	AO2.3	E1	Statement is incorrect if either matrix is singular/has determinant equal to zero as the inverse will not exist
	Correctly gives an example of a singular matrix.	AO1.1b	B1	Eg $\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$ is singular
(b)	Correctly refines the statement using 'non-singular' or equivalent wording	AO2.3	B1	Given any two non-singular square matrices, A and B , then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
(c)	Correctly recalls the inverse property for matrices A and B (seen at least once)	AO1.2	B1	A and B are non-singular so inverses exist hence $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1}$
	Correctly uses associativity by regrouping (seen at least once)	AO2.5	B1	$= \mathbf{A}\mathbf{I}\mathbf{A}^{-1}$ $= \mathbf{A}\mathbf{A}^{-1}$ $= \mathbf{I}$
	Correctly applies the identity property	AO2.1	R1	Since

throughout and concludes their rigorous mathematical argument with no errors or omissions			$(AB)(B^{-1}A^{-1}) = I$ Then $(AB)^{-1} = (B^{-1}A^{-1})$
Total 6 marks			

Q5.

(a) $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$= \begin{bmatrix} cX + sY \\ -sX + cY \end{bmatrix}$$

B1

M1

A1

3

(b) (i) $(cX + sY)^2 - 6(cX + sY)(-sX + cY) - 7(-sX + cY)^2 = 8$

$$(c^2X^2 + 2csXY + s^2Y^2) - 6([c^2 - s^2]XY) + sc\{Y^2 - X^2\}$$

$$- 7(s^2X^2 - 2csXY + c^2Y^2) = 8$$

Substn. for x & y in eqn. and multiplying out

M1

$$p = c^2 + 6sc - 7s^2$$

A1

$$q = 16cs - 6(c^2 - s^2)$$

AG

A1

$$r = s^2 - 6sc - 7c^2$$

A1

(ii) Factorising:

$$3s^2 + 8sc - 3c^2 = (3s - c)(s + 3c) = 0$$

Or by double angles

M1A1

Deducing a tan value

M1

$$\tan \theta = \frac{1}{3} \quad (\theta \text{ acute})$$

A1

$$\cos \theta = \frac{3}{\sqrt{10}}, \sin \theta = \frac{1}{\sqrt{10}}$$

Both

A1

Subst^g. sensible values back for p and r

M1

$$2X^2 - 8Y^2 = 8$$

A1

$$\frac{X^2}{2^2} - \frac{Y^2}{1^2} = 1$$

CSO

A1

8

- (iii) Since C' is a hyperbola, and it is just C rotated, it follows that C is a hyperbola

E1

1

[16]

Q6.

- (a) (i) Characteristic eqn $\lambda^2 - 9\lambda + 14 = 0$
M1A0 if num error(s) made

M1A1

$$\lambda = 2, 7$$

A1

Substituting back for at least one eval

$$\text{for } \lambda = 2, -x + 3y = 0 \text{ or}$$

$$\text{for } \lambda = 7, -2x + y = 0$$

m1

evecs $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
or non-zero multiples

A1A1

6

(ii) $\mathbf{U} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}, \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

Columns of $\mathbf{U}$ and $\mathbf{D}$ are interchangeable,
but must match;
ft wrong answers in (i)

$\mathbf{U}^{-1} = \frac{1}{\det \mathbf{U}} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$
adjoint matrix;

ft incorrect $\mathbf{U}$ (provided $\det \neq 0$)

B1FB1F

B1F

B1F

4

(b) (i) evals of $\mathbf{M}^3$ are λ^3, μ^3

B1

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since $\mathbf{M}^3 = \mathbf{U} \mathbf{D}^3 \mathbf{U}^{-1}$

E1

2

(ii) evecs of $\mathbf{M}^3$ are $\mathbf{v}_1$ and $\mathbf{v}_2$

B1

1

[13]

Q7.

(a) (i) $\frac{1}{5} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$
1/det

B1

Transposed matx. of cofactors

B1

2

(ii)
$$\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \frac{1}{5}(X + 2Y) \\ \frac{1}{5}(Y - 2X) \end{bmatrix}$$

ft

M1

A1F

2

(b)
$$\mathbf{A} = \sqrt{5} \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

B1

Enlargement sf $\sqrt{5}$ (centre O)
Two components in any order

M1A1

+ Rotation thro' $\cos^{-1}(1/\sqrt{5})$
or 63.4° or 1.11 rads

M1A1

5

(c) (i) $p = \frac{1}{2}, q = 3$ noted

$$\frac{x^2}{1} + \frac{y^2}{3} = 1$$

Or form $\frac{x^2}{2}$ shown

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B1

1

(ii) $6x^2 + y^2 = 3 \Rightarrow$

$$\frac{6}{25}(X^2 + 4XY + 4Y^2) + \frac{1}{25}(Y^2 - 4XY + 4X^2) = 3$$

Subst^g. for x and y

M1

$$\Rightarrow 10X^2 + 20XY + 25Y^2 = 75$$

$$\Rightarrow 2X^2 + 4XY + 5Y^2 = 15$$

ANSWER GIVEN

A1

2

(iii) It is just an enlarged rotation of E ,
 hence still an ellipse

B1

1

[13]

Q8.

(a) (i) $\mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$
 $\mathbf{D}, \mathbf{U}$ (alt. choices ok)

B1B1

$$\mathbf{U}^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

ft 1st B1 provided $\det \neq 0$

B1

ft 2nd B1 in non-trivial cases

B1

4

(ii) $\mathbf{M} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$
Some attempt at mtx. multn.

M1

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 12 & -3 \\ 6 & -3 \end{bmatrix} \text{ or } \begin{bmatrix} 12 & -3 \\ 6 & -3 \end{bmatrix}$$

First multn. correct ft

A1

$$\begin{aligned} & \frac{1}{2} \begin{bmatrix} 3 & -3 \\ 6 & -12 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & -3 \\ 24 & -9 \end{bmatrix} \end{aligned}$$

Ft missing $\frac{1}{2}$ only

A1

3

(b) (i) When n even, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & 3^n \end{bmatrix}$
Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ -2 \cdot 3^n & 3^n \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 3^n & 3^n \\ 2 \cdot 3^n & 4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

Showing $\mathbf{M}^n = 3^n \mathbf{I}$ legitimately

A1

3

(ii) When n odd, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & -3^n \end{bmatrix}$
 Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ 2 \cdot 3^n & -3^n \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 3^n & -3^n \\ 2 \cdot 3^n & -4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

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Showing $\mathbf{M}^n = 3^{n-1} \mathbf{M}$ legitimately

A1

3

[13]

Q9.

(a) $y = 0$ (or "x-axis") and $y = x$

$$\text{or } \mathbf{r} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{r} = b \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

B1,B1

$y = 0$ is a line of invariant points since $\lambda = 1$

Allow if proven from $(x', y') = (x, y)$ or ft
 from their line corresponding to $\lambda = 1$

B1

3

(b) (i) $\mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$
ft $\mathbf{U}$ from $\mathbf{D}$

B1,B1

2

(ii) $\mathbf{U}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
ft from $\mathbf{U}$ (provided non-singular)

B1

$\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
Attempt

M1

$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
ft first multiplication

A1

$= \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$
 CAO NMS $\Rightarrow 0$

A1

4

(iii) $\mathbf{D}^x = \begin{bmatrix} 1 & 0 \\ 0 & 2^x \end{bmatrix}$
Noted or used

B1

$\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$
*Used; **must** actually do some multiplying*

M1

$= \begin{bmatrix} 1 & 2^x - 1 \\ 0 & 2^x \end{bmatrix}$

A1

3

[12]

Q10.

(a) Char. Eqn. is $\lambda^2 - 5\lambda - 6 = 0$

B1

Solving $\Rightarrow \lambda = -1$ or 6

M1 A1

Subst^g. either λ back

M1

$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow$ evecs. $\alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
Any non-zero α

A1

$\lambda = 6 \Rightarrow 5x - 2y = 0 \Rightarrow$ evecs. $\beta \begin{bmatrix} 2 \\ 5 \end{bmatrix}$
Any non-zero β

A1

6

(b) (i) $\mathbf{D} = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}$ $\mathbf{U} = \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix}$
ft evals.

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B1F

ft evecs. (must correspond to their evals.)

B1F

2

(ii) $\mathbf{U}^{-1} = \frac{1}{7} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$

ft their $\mathbf{U}$ (provided non-singular)

B1F

1

(iii) $\mathbf{X}^5 = \mathbf{U} \mathbf{D}^5 \mathbf{U}^{-1}$

M1

$$= \frac{1}{7} \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6^5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$$

Use of correct $\mathbf{D}^5$ (ft)

N.B. $6^5 = 7776$

B1F

$$= \begin{bmatrix} 2221 & 2222 \\ 5555 & 5554 \end{bmatrix}$$

A1

3

[12]

Q11.

(a) $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 4+k \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ when $k = 5$,
 $M \times$ given evec.

ft

$$\lambda = 9$$

M1A1

A1

3

(b) Char. Eqn. is $\lambda^2 - 7\lambda - 18 = 0$

M1A1

$(\lambda - 9)(\lambda + 2) = 0$ and 2nd eval. is -2

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A1

OR $\det M = \lambda_1 \lambda_2 \Rightarrow -18 = 9\lambda_2$
 $\Rightarrow \lambda_2 = -2$

Or via trace $M = \lambda_1 + \lambda_2$

Substg. $\lambda = -2 \Rightarrow 4x + 7y = 0$

M1

$\Rightarrow$ evec. $\begin{bmatrix} 7 \\ -4 \end{bmatrix}$

A1

5

(c) $D = \begin{bmatrix} -2 & 0 \\ 0 & 9 \end{bmatrix}$, $U = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix}$
 ft (alternatives possible)

B1B1

2

(d) $U^{-1} = \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$
ft non-trivial U's

B1

1

(e) $M^{2n} = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} (-2)^{2n} & 0 \\ 0 & (9)^{2n} \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 7 \times 4^n & 81^n \\ -4^{n+1} & 81^n \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$
For D^{2n}

B1

or

$\begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 4^n & -4^n \\ 4 \times 81^n & 7 \times 81^n \end{bmatrix}$
CAO any correct form

M1
A1

Thus $a = \frac{1}{11} \{7 \times 4^n + 4 \times 81^n\}$
i.e. $p = \frac{7}{11}, q = \frac{4}{11}$

A1

4

In its original form, the question asked for the following conclusion to be made:

Since a is an integer, and $\text{hcf}(4, 11) = 1$,
 $7 \times 4^{n-1} + 81^n$ is a multiple of 11

[15]

Examiner reports

Q1.

Only half of the students were able to make satisfactory progress with this question. Most

of the successful attempts substituted $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (or equivalent) for **A**, and then solved the resulting simultaneous equations. Only the most able students were able to successfully isolate **A** or $\mathbf{A}^{-1}$ by rearranging the given matrix equation.

Q5.

Although there were a few excellent, complete responses to this question, in general it turned out to reflect the lack of preparedness on the part of candidates for the demands presented to them here. The three marks in part (a) were readily and easily acquired by most candidates attempting it. However, 156 candidates were unable to proceed beyond the given result of part (a), and made no attempt at the remaining parts of this question.

In part (b)(i), it was not uncommon to find that candidates were less than comfortable with squaring out their x and y terms in the new variables X and Y , and collecting them up proved to be far more problematical than should have been the case. Rather a lot of candidates seemed to think that the XY term came exclusively from the xy term, and such efforts foundered at this point.

Few candidates got to the correct equation of the hyperbola. A suitable answer to part (b)(iii) required only an awareness that the original curve had been rotated into the final hyperbola, but only a few candidates answered it without having worked their way to it. Of these, most offered spurious reasons, such as the not uncommon reference to the suggestion that the original curve was a hyperbola because this final one was an ellipse.

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Q6.

Candidates generally showed a very good grasp of eigenvalues and eigenvectors in part (a)(i), though some were confused when forming the eigenvectors. The responses to part (a)(ii) were almost equally successful, but some faltered in tackling the inverse of a 2 by 2 matrix, either failing to include the reciprocal of the determinant or incorrectly manipulating the elements of the matrix to form the adjoint matrix.

In part (b)(i), many candidates earned partial credit, either by giving the eigenvalues but failing to show the connection with the diagonalised form, or by doing some good work with the diagonalised form and then failing to state clearly what the eigenvalues were. Part (b)(ii) was answered correctly in many cases but a common error was to write down the meaningless answers ' $\mathbf{v}_1^3$ ' and ' $\mathbf{v}_2^3$ '. Some candidates used the matrix from part (a) in their answers to part (b), but this was accepted as they could still show whether they understood what was needed in part (b).

Q7.

It was disappointing to see so many candidates unable to find correctly the inverse of a 2×2 matrix at the outset of this question. The description in part (b) proved tricky — as had been expected — but actually only relies on understanding of the FP1 work on matrices and transformations. Taking this into account, it was sad to see so few attempts to describe **T**.

In part (c)(i), many candidates failed to point out the values of p and q that would justify E 's status as an ellipse. In most cases, this arose as a result of an inability to divide correctly throughout the given equation by 3. By the time it came to the final two parts, most candidates had made at least one mistake which deprived them of the chance of further successful progress.

Q8.

The more confident candidates found this question to be a source of easy marks. The real difficulties arose in part (b) due to a widespread inability to use brackets and deal with minus signs. When candidates wrote -3^n when they actually meant $(-3)^n$, and it was necessary to read on to see what they eventually did with the terms involving powers of (-3) . The standard approach appears in the mark scheme, though the alternative algebraic ones also appeared quite often, and these were much more pleasing to the eye. For n even, noting that $\mathbf{D}^n = 3^n \mathbf{I}$, we have $\mathbf{M}^n = 3^n \mathbf{U} \mathbf{I} \mathbf{U}^{-1}$, which gives the required result both obviously and quickly. However, a small number attempted this approach, but wrote something along the lines of $\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1} = \mathbf{D}^n \mathbf{U} \mathbf{U}^{-1} = \mathbf{D}^n$ first. This is wrong and

scored 0/3. For n odd, a similar 'shortcut' arises via $\mathbf{M}^n = 3^n \mathbf{U} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{U}^{-1}$. However, a very small number of candidates are to be highly commended for the following insightful, approach: if n is odd, then $n - 1$ is even, so that $\mathbf{M}^n = \mathbf{M}^{n-1} \mathbf{M} = (3^{n-1} \mathbf{I})(\mathbf{M})$ using the result of part (b)(i) $= 3^{n-1} \mathbf{M}$, as required.

Q9.

Most difficulties with this question arose in part (a), where candidates had to consider the meaning and relevance of the information given to them in the question. The most commonly lost mark was in the widespread failure to point out that the line of invariant points was flagged up by the fact that the eigenvalue corresponding to the x -axis was 1. In general, most other marks lost on this question arose as a result of minor slips and oversights in the working. A few candidates didn't seem to know how to find $\mathbf{M}^n$.

Q10.

This question was handled very confidently by most candidates, and apart from a small number of candidates who clearly deduced the matrix $\mathbf{X}^5$ from a calculator, despite the question's injunction to use the previous results, marks of 10, 11 or 12 were very common indeed.

Q11.

Most candidates managed to score some marks here, Even those picking up 2 or 3 of the marks available here did so retrospectively. The most common error in part (b) was to fail to turn the equation $4x + 7y = 0$, gained from the second eigenvalue of -2 , into a correct eigenvector. Also, the factor of $1/\det \mathbf{U}$ was often overlooked. Part (e) was seldom attempted. It was, however, very pleasing to see some completely correct conclusions to the paper.